GBM Confidence-Enhanced Survival Inference Project

Student Guide with Instructions and Sample Output Figures

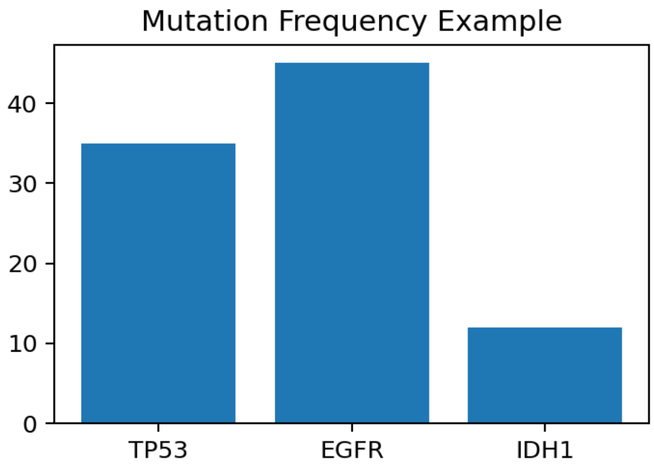
Step 1: Cohort Characterization

Clinical Question: Is our GBM cohort representative enough to support reliable genomic survival analysis?

Student Instructions: Load the dataset, remove patients missing survival outcomes, summarize demographics, mutation frequencies, and missingness. Create Table 1 and compare mutation frequencies with published GBM studies.

Expected Output: Table 1, mutation frequency chart.

Example Interpretation: The cohort appears representative of typical GBM populations.



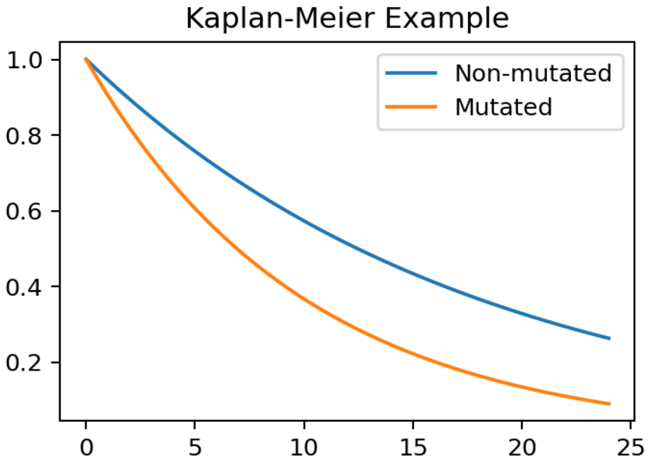
Step 2: Single-Mutation Survival Analysis

Clinical Question: Which individual mutations are associated with clinically meaningful differences in survival?

Student Instructions: For each mutation, separate patients into mutated and non-mutated groups, generate Kaplan-Meier curves, compute median survival and log-rank p-values.

Expected Output: Kaplan-Meier plots and survival summary table.

Example Interpretation: TP53-mutated patients show shorter survival.



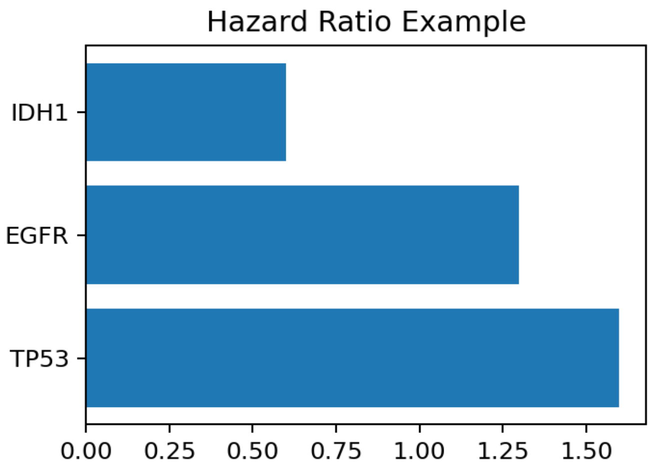
Step 3: Adjusted Cox Survival Modeling

Clinical Question: Which mutations provide independent prognostic information?

Student Instructions: Fit unadjusted and adjusted Cox models using age, sex, and available covariates. Create a forest plot and compare adjusted vs unadjusted hazard ratios.

Expected Output: Hazard ratio table and forest plot.

Example Interpretation: TP53 remains associated with increased mortality after adjustment.



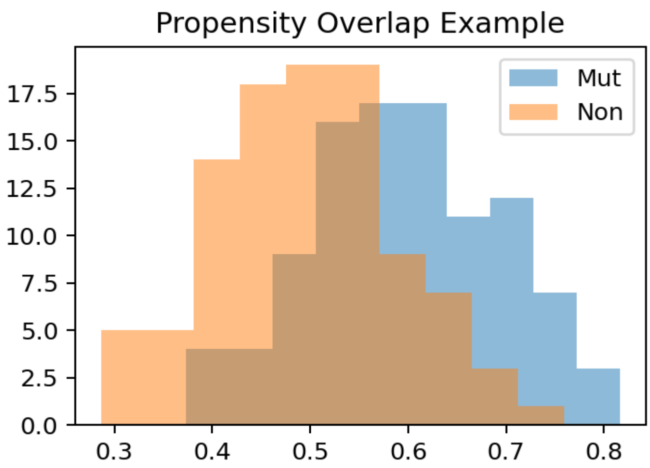
Step 4: Propensity Score Enhanced Analysis

Clinical Question: Are mutation-survival associations robust after balancing patient characteristics?

Student Instructions: Estimate propensity scores, evaluate overlap, compute IPTW weights, and fit weighted Cox models. Compare results with Step 3.

Expected Output: Overlap plots and weighted hazard ratios.

Example Interpretation: The association remains after balancing measured covariates.



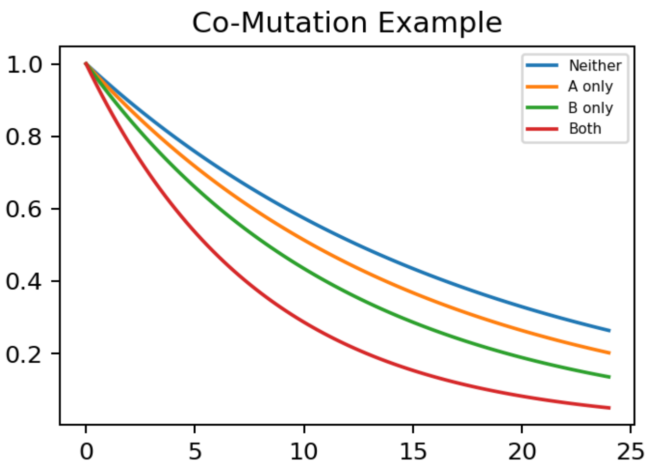
Step 5: Co-Mutation Analysis

Clinical Question: Do combinations of mutations define distinct molecular survival phenotypes?

Student Instructions: Create four groups (neither, A only, B only, both) and analyze TP53+EGFR, TP53+IDH1, ATRX+IDH1, and TERT+EGFR.

Expected Output: Four-group survival curves and co-mutation HR table.

Example Interpretation: Co-mutated patients have the worst survival.



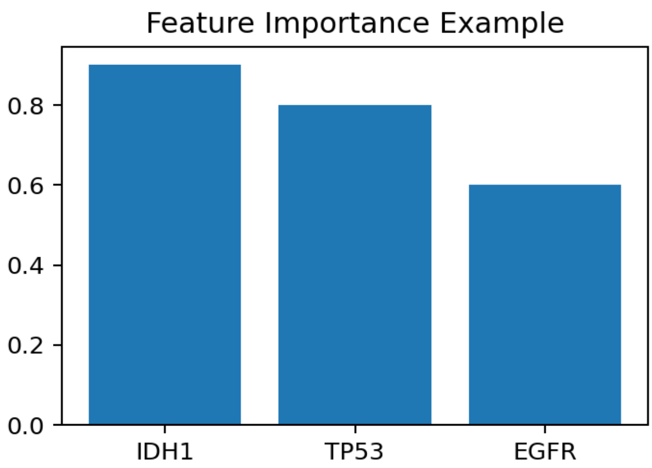
Step 6: Survival Machine Learning

Clinical Question: Which mutations contribute most strongly to survival prediction?

Student Instructions: Build Random Survival Forest and optionally XGBoost Survival. Evaluate with C-index and compute feature importance or SHAP values.

Expected Output: Model performance table and mutation ranking plot.

Example Interpretation: IDH1 is a strong survival biomarker.



Step 7: Confidence Framework

Clinical Question: Which biomarkers are sufficiently reproducible to be trusted clinically?

Student Instructions: Combine evidence from KM, Cox, IPTW, and ML analyses. Assign High, Moderate, or Low confidence and rank biomarkers.

Expected Output: Confidence ranking table.

Example Interpretation: TP53 and IDH1 emerge as high-confidence biomarkers.